MCKV Institute of Engineering

243 G. T. Road (N), Liluah, Howrah – 711204

Subject: **Database Management System Lab** Code:**PC-CS692**
Stream: CSE(DS) Credit: 2

Assignment–1

Create the following tables using the schema given below and insert given data set accordingly.

Table Name- client _master
Description- Used to store client information

Column No	Column Name	Data Type	Size	Attributes
1	Client_no	Varchar2	6	Primary key, first letter must start with 'C'
2	Name	Varchar2	30	Not NULL
3	Address1	Varchar2	30	
4	Address2	Varchar2	30	
5	City	Varchar2	15	
6	State	Varchar2	15	
7	Pincode	Number	6	
8	Balance_due	Number	10,2	

Data of client_master table

Col-1	Col-2	Col-3	Col-4	Col-5	Col-6	Col-7	Col-8
C001	Ivan Bayross	P-76	Worli	Bombay	Maharastra	400054	15000
C002	VandanaSatiwal	128	Adams Street	Madras	TamilNadu	780001	0
C003	PramadaJaguste	157	Gopalpur	Kolkata	West Bengal	700058	5000
C004	BasuNavindgi	A/12	Nariman	Bombay	Maharastra	400056	0
C005	Ravi Sreedharan	B/34	Rajnagar	Delhi	Delhi	100001	2000
C006	Rukmini	Q-12	Bandra	Bombay	Maharastra	400050	0

Table Name- product_master:
Description- Used to store product information

Column No	Column Name	Data Type	Size	Attributes
1	Product_no	Varchar2	6	Primary key, First letter must start with 'P'
2	Description	Varchar2	40	Not null
3	Profit_percent	Number	4,2	Not null
4	Unit_measure	Varchar2	10	Not null
5	Qty_on_hand	Number	8	Not null
6	Reorder_level	Number	8	Not null
7	Sell_price	Number	8,2	Not null, cannot be 0
8	Cost_price	Number	8,2	Not null, cannot be 0

Data of product_master table

Col-1	Col-2	Col-3	Col-4	Col-5	Col-6	Col-7	Col-8
P00001	1.44 Floppies	5	Piece	100	20	525	500
P03453	Monitors	6	Piece	10	3	12000	11280
P06734	Mouse	5	Piece	20	5	1050	1000
P07865	1.22 Floppies	5	Piece	100	20	525	500
P07868	Keyboard	2	Piece	10	3	3150	3050
P07885	CD Drive	2.5	Piece	10	3	5250	5100
P07965	540 HDD	4	Piece	10	3	8400	8000
P07975	1.44 Drive	5	Piece	10	3	1050	900
P08865	1.22 Drive	5	Piece	2	3	1025	850

Table Name- salesman_master:
Description- Used to store salesman working for company

Column No	Column Name	Data Type	Size	Attributes
1	Salesman_no	Varchar2	6	Primary key,first letter must start with 'S'
2	Salesman _name	Varchar2	30	Not null
3	Address1	Varchar2	30	Not null
4	Address2	Varchar2	30	
5	City	Varchar2	20	
6	Pincode	Number	8	
7	State	Varchar2	20	
8	Sal_amt	Number	8, 2	Not null, cannot be 0

Data of salesman_master table

Col-1	Col-2	Col-3	Col-4	Col-5	Col-6	Col-7	Col-8
S001	Kiran	A/14	Worli	Bombay	400002	Maharastra	3000
S002	Manish	65	Nariman	Bombay	400001	Maharastra	3000
S003	Ravi	P-7	Bandra	Bombay	400032	Maharastra	3000
S004	Asish	A/5	Juhu	Bombay	400044	Maharastra	3000

Table Name- sales_order:
Description- Used to store client's orders

Column No	Column Name	Data Type	Size	Attributes
1	Order_no	Varchar2	6	Primary key, first letter must start with 'O'
2	Order_date	Date		
3	Client_no	Varchar2	6	Foreign key references Client_master table
4	Salesman_no	Varchar2	6	Foreign key references salesman _master table
5	Delivery_type	Char	1	Delivery part(P),full(F) Default 'F'
6	Bill_y_n	Char	1	
7	Delivery_date	Date		Cannot be less than Order_date
8	Order_status	Varchar2	10	Values('InProcess',' Fullfilled', 'BackOrder', 'Cancelled')

Data of sales_order table

Col-1	Col-2	Col-3	Col-4	Col-5	Col-6	Col-7	Col-8
O19001	12-Jan-96	C001	S001	F	N	20-Jan-96	InProcess
O19002	25-Jan-96	C002	S002	P	N	27-Jan-96	BackOrder
O46865	18-Feb-96	C003	S003	F	Y	20-Feb-96	Fullfilled
O19003	03-Apr-96	C001	S001	F	Y	07-Apr-96	Fullfilled
O46866	20-May-96	C004	S002	P	N	22-May-96	Cancelled
O19008	24-May-96	C005	S004	F	N	26-May-96	InProcess

Table Name- sales_order_details:
Description- Used to store client's orders with details of each product ordered

Column No	Column Name	Data Type	Size	Attributes
1	Order _no	Varchar2	6	Foreign key referencessales_order table
2	Product_no	Varchar2	6	Foreign key references product_master table
3	Qty_ordered	Number	8	
4	Qty_disp	Number	8	
5	Product_rate	Number	10, 2	

Data of sales_order_details

Col-1	Col-2	Col-3	Col-4	Col-5
O19001	P00001	4	4	525
O19001	P07965	2	1	8400
O19001	P07885	2	1	5250
O19002	P00001	10	0	525
O46865	P07868	3	3	3150
O46865	P07885	3	1	5250
O46865	P00001	10	10	525
O46865	P03453	4	4	1050
O19003	P03453	2	2	1050
O19003	P06734	1	1	12000
O46866	P07965	1	0	8400
O46866	P07975	1	0	1050
O19008	P00001	10	5	525
O19008	P07975	5	3	1050

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Assignment- 2

1. Find the names of all clients having ‘a’ as the second letter in their names.
2. Find out the clients who do not stay in a city whose first letter is ‘B’.
3. List the names and city of all clients who have exactly 12 characters in length and starts with ‘I’.
4. Find the list of all clients who stay in ‘Bombay’ or ‘Delhi’.
5. Print the list of all clients whose bal_due is greater than value 10,000.
6. Print the information from sales_order table for orders places in the month of January.
7. Display the order information for client_no ‘C001’ and ‘C002’.
8. Find products whose selling price greater than 2000 and less than 5000.
9. Find products whose selling price is more than 1500. Calculate a new selling price as original sellingprice*1.15. Rename the new column in the above query is New_price.
10. List the names, city and state of clients who are not in the state of ‘Maharastra’.
11. Display the month(in alphabets) and date when the order must be delivered.
12. Display the Order_date in the format ‘DD-Month-YY’ e.g 12-February-13.
13. Find the date, 15 days after today’s date.

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Assignment-3

1. Count the total number of orders.
2. Calculate the average price of all the products.
3. Count the number of products having price greater than or equal to 1500.
4. Determine the maximum and minimum product prices. Rename the output as max_price and min_price respectively.
5. Change the City of the Client_no ‘C005’ to ‘Madras’.
6. Change the Bal_due of Client_no ‘C005’ to Rs.3000/-.
7. Delete from client_master where the column state holds the value ‘Tamilnadu’.
8. Add a column called ‘Telephone’ of data type ‘number’ and size 10 in the table client_master.
9. Change the size of data type Pin_code to 10 in the table client_master.
10. Drop the column Address2 from the table client_master.
11. Create another table client_master_duplicate with the same structure of client_master(without copying the data of the table client_master).
12. Insert the data into client_master_duplicate table from client_master table.
13. Rename the table client_master_duplicate to c_master.
14. Destroy the table c_master with its data.

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Assignment-4

Do the following experiment with the default tables of oracle such as ***EMP, DEPT*** *etc.*

1. Display the names of all employees' right aligning them to 15 characters.
2. Display the names of all employees' padding them to the right up to 15 characters with '*'.
3. Find the details of all the managers in department 10 and all clerks in department 20 and all employees who are neither managers nor clerks but whose salary is more than or equal to 2000/-.
4. List all the employees who have joined between 01/02/81 and 31/08/81.
5. List all the employees who were joined as manager during 1981.
6. List the employees whose salaries are 800, 1600 or 2450.
7. List the names of all employees who are either 'clerkes' or 'salesman' or 'analyst'.
8. List the total number of employees and the average salaries of the different departments.
9. Calculate the average salary of all employees whose department is 30.
10. Calculate the minimum salary earn by a 'clerks'.
11. Calculate the maximum salary earn by salesman.
12. Find the names of those employees whose immediate boss is in different department.
13. Calculate the no of employees who are not getting any commission.
14. Find the department is not having any employee.

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Assignment-5

1. List all the employee names, dept name and the city, in department name order.
2. List all employees working in Dallas in descending order of salary.
3. List employee name, department name, job and location of all employees who work in DALLAS.
4. List the employee name, salary, PF, HRA, DA and gross salary; order the result in ascending order of gross. PF is 10% of salary, HRA is 60% of salary and DA is 40% of salary.
5. Display names and salary of all the employees who report to KING.
6. List all employees who work in DALLAS and earn more than any employee working in Chicago.
7. List all employees who work in the same post as Smith.
8. Find the job with the highest average salary.
9. List the top 10 earners in the company.
10. Display the names of all employees' replacing 'A' with 'a'.
11. Show the salary of all the employees rounding it to the nearest Rs.1000/-.
12. Show the first three and last three characters of the names of all the employees.

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Assignment-6

Table: Client_master

Column_Name	Data type	Size	Attributes
Client_no	Varchar2	8	Primary Key
Name	Varchar2	20	Not Null
Address1	Varchar2	20	Not Null
Address2	Varchar2	20	
City	Varchar2	15	
State	Varchar2	15	
Pincode	Varchar2	8	
Bal_due	Number	8,3	

1. Create a view vw_client_master using Client_no, Name, Address1 and Bal_due
   a. Insert at least 3 records to vw_client_master.
   b. Update a record to vw_client_master.
   c. Delete a record from vw_client_master.

And check that the above operation if the base table is affected or not.

2. Create a view Vw_sales_det using Client_no, Order_no, Order_date, Product_no, Qty_ordered, and order_status for all order which have already marked as 'Backorder' .(Using the tables sales_order, sales_order_details).
   a. Insert a record to vw_sales_det.
   b. Update the client_no for a particular order_no.
   c. Delete a record.
   d. Remove the views from database.

Assignment-7

Write PL/SQL code Blocks for the following.

1. Write a PL/SQL code for finding factorial of a given number
2. Write a PL/SQL code for calculating finding the sum of N numbers.
3. Write a PL/SQL code for finds a given year is leap year or not.
4. Write a PL/SQL code for finding maximum of three numbers. (Input will be given by the user).
5. Write a PL/SQL code block to calculate the area of a circle for a value of radius varying from 6 to 10. Store the radius and corresponding values of calculated area in an empty table named Areas, Consisting of two columns Radius and Area.
6. Write a PL/SQL code block that will accept a client_no from the user and adds the amount of Rs. 1000 to bal_due column, has a minimum balance of Rs. 6000. The process is fire on client_master.

Assignment-8

1. a) Create a table whose structure will be as follows:

Table Name: Prime_Entry

Column Name	Data Type	Attributes
Num_id	Number(3)	Primary Key
Prime_num	Number(3)	Not Null

b) Write a PL/SQL block of code that will take a number from user and test whether the number is prime or not. If the number is prime, then enter into above table by generating NUMID automatically.
c) Now add a checking for same prime number entry. It will show - 'Number already exists in database' for same prime number entry. Write a function to test whether given number exist or not.

2. Create the following table:

Table Name: Acc_details

Column_Name	Data type	Size	Attributes
Acc_no	Varchar2	8	Primary Key
Name	Varchar2	20	Not Null
Address	Varchar2	20	Not Null
DOB	Date		Not Null
Sex	Char	1	Not Null, Values ('M', 'F')
Contact_no	Number	10	Not Null
Last_trans_date	Date		Not Null
Total_amt	Number	12,4	Not Null
Acc_status	Char	1	Not Null, Values ('A', 'I')

Table Name: Transactions_Acc

Column_Name	Data type	Size	Attributes
Transaction_id	Number	8	Primary Key
Acc_no	Number	8	References Acc_details.Acc_no
Deposit_amt	Number	12,4	
Withdraw_amt	Number	12,4	
Mode_trans	Char	5	Not Null
Cheque_no	Number	6	Default 0
Trans_date	Date		Not Null

When a specific account will be deleted then all the transaction details from Transactions_acc will be deleted for that account number.

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Assignment-9

1. Write a PL/SQL block of code that first withdraws an amount of Rs. 500. Then again withdraws Rs. 500. Now if the current balance of a specific account number is less than Rs. 1000 then undo the last withdraw just made.

2. Write a PL/SQL block of code to update the location of specific department number that will be taken from user. Display an appropriate message using SQL%FOUND based on existence of the record in the Department table and display an appropriate message using SQL%NOTFOUND based on the non-existence of the record in Department Table.

3. Write a PL/SQL block that will show an Employee name for a given Employee number. Here you try to enter a wrong Employee number and show an appropriate message i.e. NOT FOUND using exception handling.

4. Write a PL/SQL block of code using your own exception handling that will show an error message whenever you want to insert a null value in a not null column.

5. a) Create a table Emp_sal_inc that have three column (Emp_id, Cur_sal, Inc_date).
   b) Now write a PL/SQL block of code will allow 2% salary increment of all employee of RESEARCH department. After that all records are to be inserted into the above table i.e. Emp_sal_inc.

Asssignment-10

1. Write a PL/SQL block that will add 2% interest of all customer of a bank for active account.
   i) For updating Acc_details updating, you have to use Cursor.
   ii) For entry in Transaction_Acc, you have to use procedure.
   iii) For Generation Transaction_id, you have to use function.

2. a) Create the following table: (Table Name:– Emp_audit)

Column_Name	Data type	Size	Attributes
Emp_no	Number	4	Primary Key
Dept_no	Number	4	Not Null, Ref. department.dept_no
Status	Varchar 2	8	
Salary	Number	8,2	Not Null
Audit_date	Date		Not Null

b) Write a trigger that must keep track of records (in above table) that are being deleted or updated from Employee table.

c) Write a SQL command to update the employee entry and describe the output.

1.

2.

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Signatures of the Faculty Members Signatures of HOD - CSE)